

Network Intrusion Detection Using Random Forest and Deep Neural Networks

MACHINE LEARNING

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ABSTRACT

Network Intrusion Detection Using Random Forest and Deep Neural Networks is a machine learning-based cybersecurity system designed to detect and classify malicious network activities. The system analyzes network traffic data and identifies whether the traffic is normal or suspicious. The collected data is first preprocessed by removing unnecessary information and extracting important network features. Two machine learning approaches, Random Forest and Deep Neural Network (DNN), are used to analyze the processed data and detect different types of network attacks. Random Forest provides efficient classification based on multiple decision trees, while the DNN learns complex patterns in network traffic. The detected results, including the source IP, attack type, detection model, and status, can be stored and displayed through a dashboard. The proposed system helps in monitoring network activity, identifying potential threats, and providing useful information for faster security analysis.

INTRODUCTION

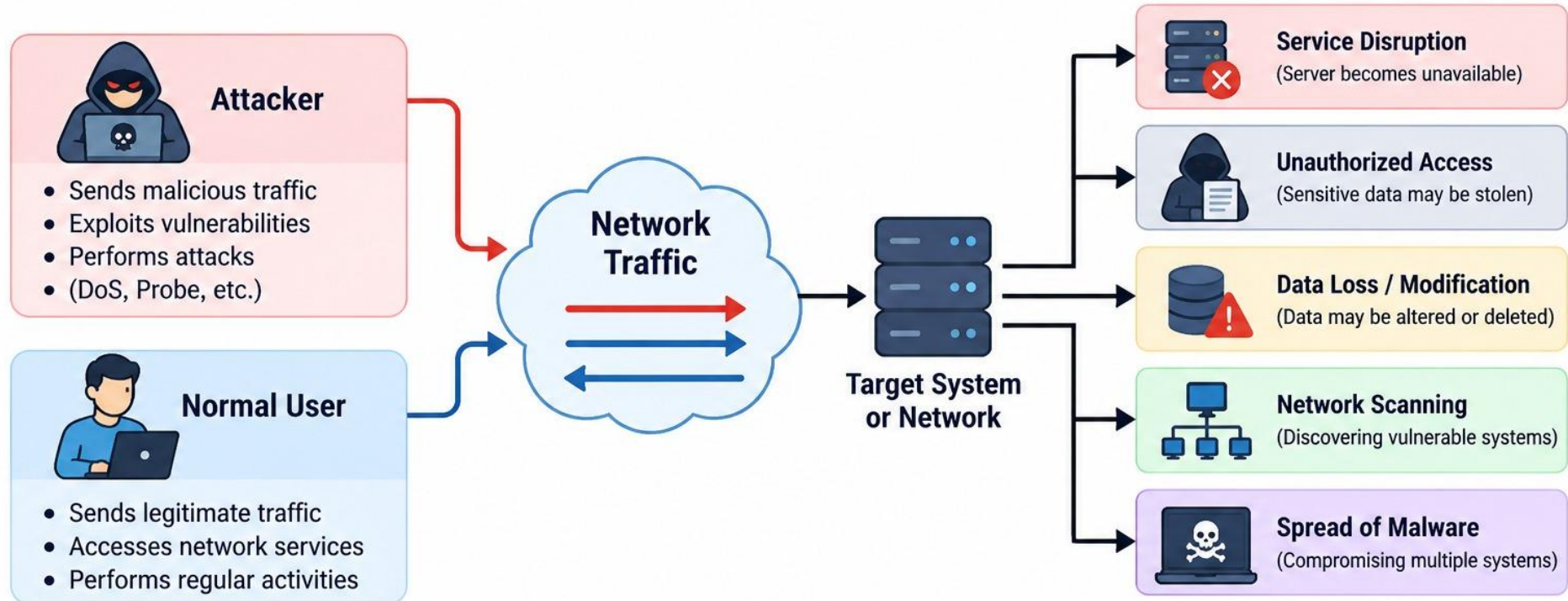
With the rapid growth of computer networks, cyber attacks and unauthorized access have become major security threats. Network Intrusion Detection Systems (NIDS) help identify suspicious activities by continuously monitoring network traffic.

This project uses Random Forest and Deep Neural Networks (DNN) to analyze network traffic and detect malicious activities. The system classifies network traffic as normal or malicious based on different network features.

Random Forest helps in accurate and efficient classification, while DNN learns complex patterns from network data. The detected threats can be stored in a database and displayed through a security dashboard.

The system can identify different types of attacks and provide useful information such as the source IP, attack type, and detection status. This helps in early threat detection, network monitoring, and improving overall cybersecurity.

Problem Statement



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With the increasing growth of network usage, computer networks are becoming more vulnerable to various cyberattacks such as **Denial of Service (DoS)**, **Probing**, **Remote-to-Local (R2L)** and **User-to-Root (U2R)** attacks. These malicious activities can lead to service disruption, unauthorized access, data loss and compromise of network security. Therefore, there is a need for an **efficient and accurate intrusion detection system** that can automatically analyze network traffic and **identify malicious activities** in real time.

Software Requirements

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VS Code – To write and run the project code

Python – Main programming language

Pandas – To handle the network dataset

NumPy – For numerical calculations

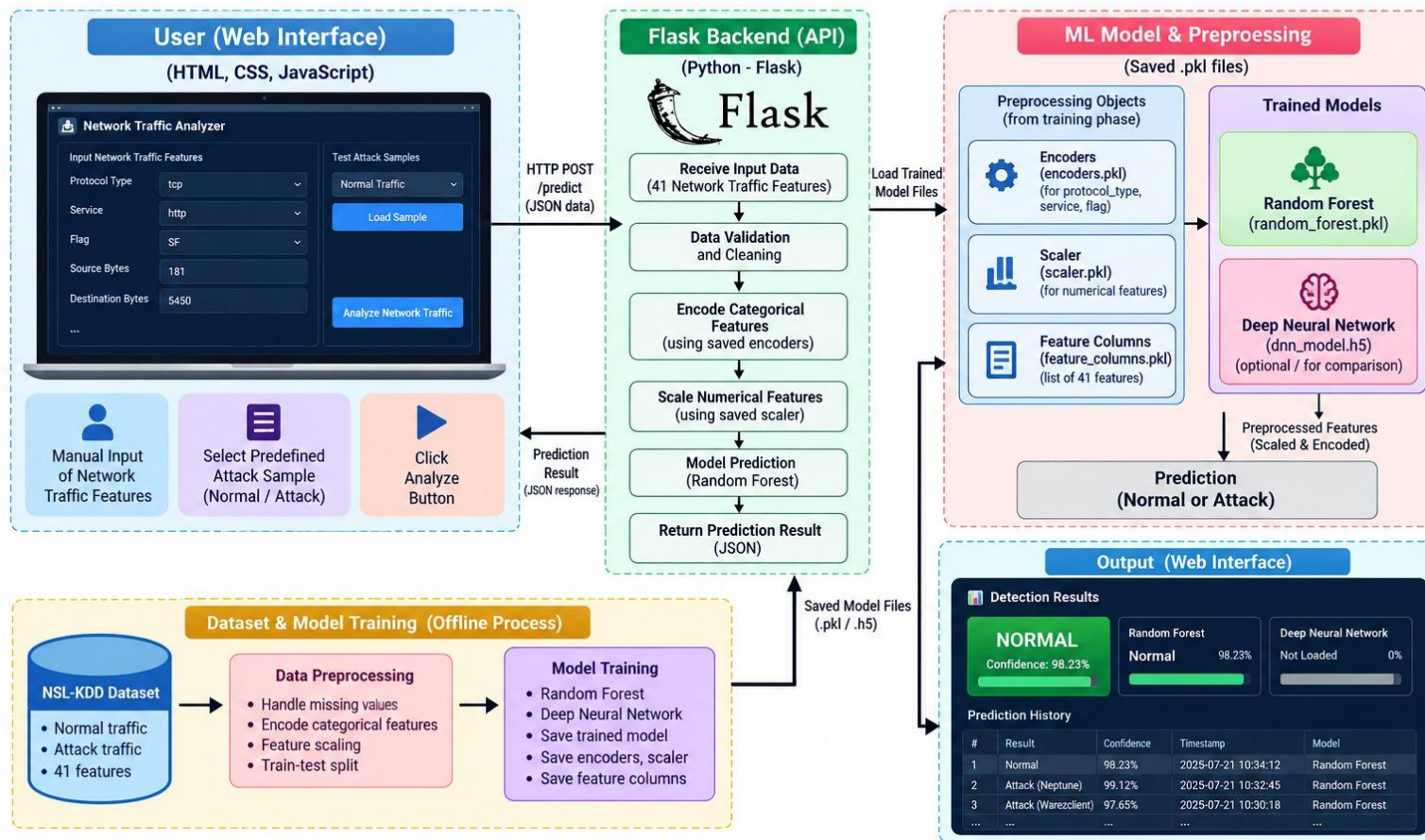
Scikit-learn – To build the Random Forest model

TensorFlow/Keras – To build the Deep Neural Network (DNN)

Matplotlib – To create graphs and visualize results

System Architecture

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Conclusion

NetworkGuard AI provides an effective machine-learning-based approach for detecting network intrusions by analyzing network traffic patterns. Using the NSL-KDD dataset, the system processes network features, applies preprocessing techniques, and uses machine-learning models to classify traffic as Normal or Attack. The Flask backend connects the trained models with a user-friendly web interface, allowing users to test different attack samples and view prediction results with confidence. Overall, the project demonstrates how machine learning can support automated network security and intrusion detection.

FUTURE SCOPE

Future Scope

The system can be further improved by integrating real-time network traffic capture instead of relying only on dataset samples. Future versions can use larger and more recent cybersecurity datasets, advanced deep-learning and hybrid models, and continuous model retraining to detect new and evolving attacks. The system could also include real-time alerts, attack history and analytics dashboards, automated threat response, cloud deployment, and integration with enterprise network monitoring tools, making it more suitable for practical cybersecurity environments.

